

Data modeling: CSCI E-106

Chapter 3 – Applied Linear Statistical Linear Model
Chapter 8 and 9 – Linear Statistical Models with R

Diagnostic and Remedial Measures

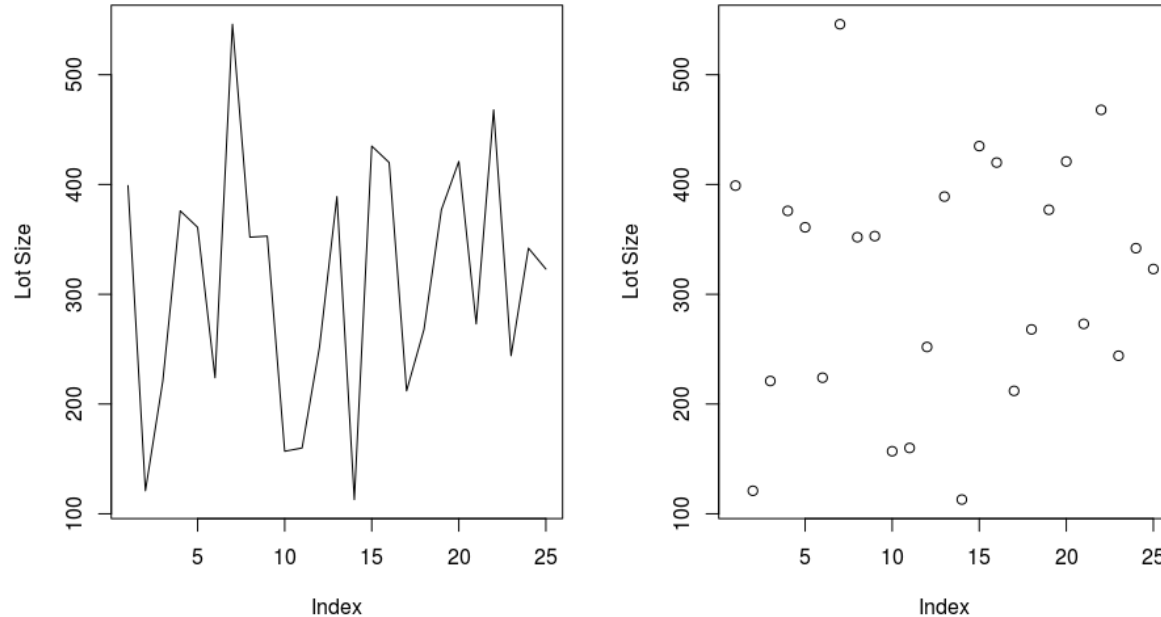
- Is the regression model **appropriate** for that application?
- It is important to examine the aptness of the model for the data before inferences based on that model are undertaken.
 - Graphs
 - Univariate analyses
 - Statistical tests
- Consider remedial techniques: when the data are not in accordance with the conditions of the regression model

Graphic diagnostics for the predictor variable

- Are there any outlying X values that could influence the appropriateness of the fitted regression function?
- Scatter Plot
- Box Plot
- Time Sequence Plot

Graphic diagnostics for the predictor variable, cont'd

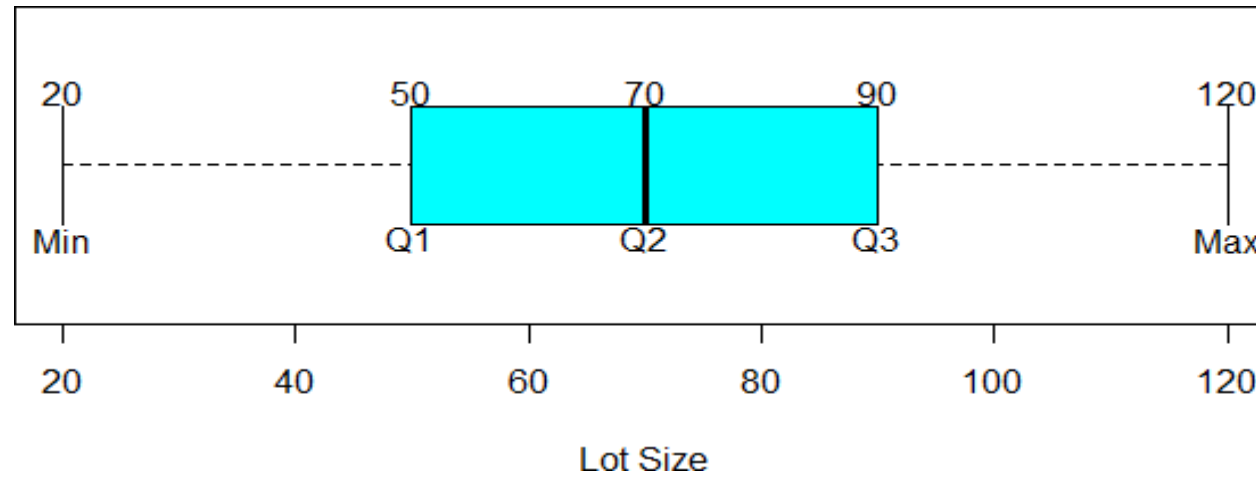
- Scatter Plot



- `par(mfrow=c(1,2))`
- `plot(toluca_data$workhrs, type="l", ylab="Lot Size")`
- `plotplot(toluca_data$workhrs, type="p", ylab="Lot Size")`

Graphic diagnostics for the predictor variable, cont'd

Boxplot



- `boxplot(Size, horizontal = TRUE, staplewex = 1,col="cyan", xlab="Lot Size")`
- `text(x=fivenum(Size), labels =fivenum(Size), y=1.25)` `text(x=fivenum(Size), labels =c("Min","Q1","Q2","Q3","Max"), y=0.75)`

Graphic diagnostics for the predictor variable, cont'd

- A time sequence plot: **production run** (time sequence) vs. **X**
- Sequence plots should be utilized whenever data are obtained in a sequence.

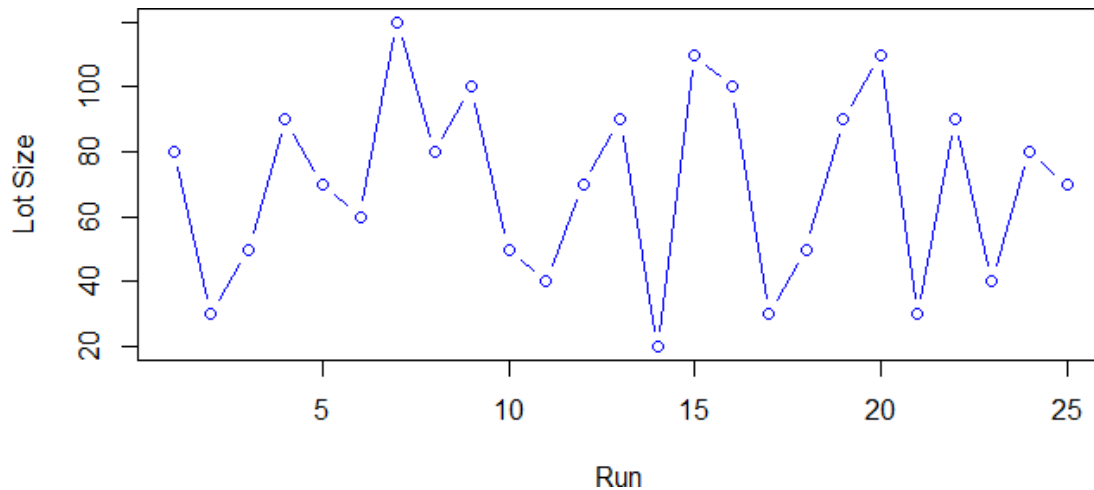


Figure : Sequence Plot for Predictor Variable-Toluca Company Example.

- Figure contains no special pattern.
- If the plot had shown that smaller lot sizes had been utilized early on and larger lot sizes later on, this information could be very helpful for subsequent diagnostic studies of the aptness of the fitted regression model.

Residuals

- Diagnostics for the response: Examination of the residuals
- Residual $\mathbf{e}_i$: $\mathbf{e}_i = Y_i - \hat{Y}_i$
- Unknown true error $\boldsymbol{\varepsilon}_i$: $\boldsymbol{\varepsilon}_i = Y_i - E\{Y_i\}$

Properties of Residuals

- Mean:

$$\bar{e} = \frac{\sum e_i}{n} = 0$$

- Variance:

$$s^2 = \frac{\sum (e_i - \bar{e})^2}{n - 2} = \frac{\sum e_i^2}{n - 2} = \frac{SSE}{n - 2} = MSE$$

If the model is appropriate, MSE is an unbiased estimator of the variance of the error terms σ^2

Properties of Residuals, cont'd

- Nonindependence: e_i are not independent since they involve $\hat{Y}_i$ which are based on the same fitted regression
- Subject to two constraints for the regression model (2.1)
 - $\sum e_i = 0$
 - $\sum X_i e_i = 0$
- When the sample size is large in comparison to the number of parameters in the regression model, the dependency effect among e_i is relatively unimportant and can be ignored.

Semistudentized Residuals

- At times, it is helpful to standardize the residuals for residual analysis. Since the standard deviation of the error terms ε_i is estimated by $\sqrt{MSE}$
- it is natural to consider the following form of standardization:

$$e_i^* = \frac{e_i - \bar{e}}{\sqrt{MSE}} = \frac{e_i}{\sqrt{MSE}}$$

- e_i^* is called a **semistudentized residual**.
 - The standard deviation of e_i is complex and varies for the different residuals e_i , and. $\sqrt{MSE}$ is only **an approximation** of the standard deviation of e_i .

Departures from Model to Be Studied by Residuals

- Using residuals for examining six important types of departures from model with normal error:

1. Not linear
2. The error terms do not have constant variance.
3. The error terms are not independent.
4. The error terms are not normally distributed.
5. The model fits all but one or a few outlier observations.
6. Important predictor variables have been omitted from the model.

formal diagnostic plots for residuals

- 1 e_i vs. X
- 2 $|e_i|$ or e_i^2 vs. X
- 3 e_i vs. $\hat{Y}_i$
- 4 e_i vs. time or other sequence
- 5 e_i vs. omitted predictor variables
- 6 Box plot of e_i
- 7 Normal probability plot of e_i .

formal diagnostic plots for residuals, cont'd

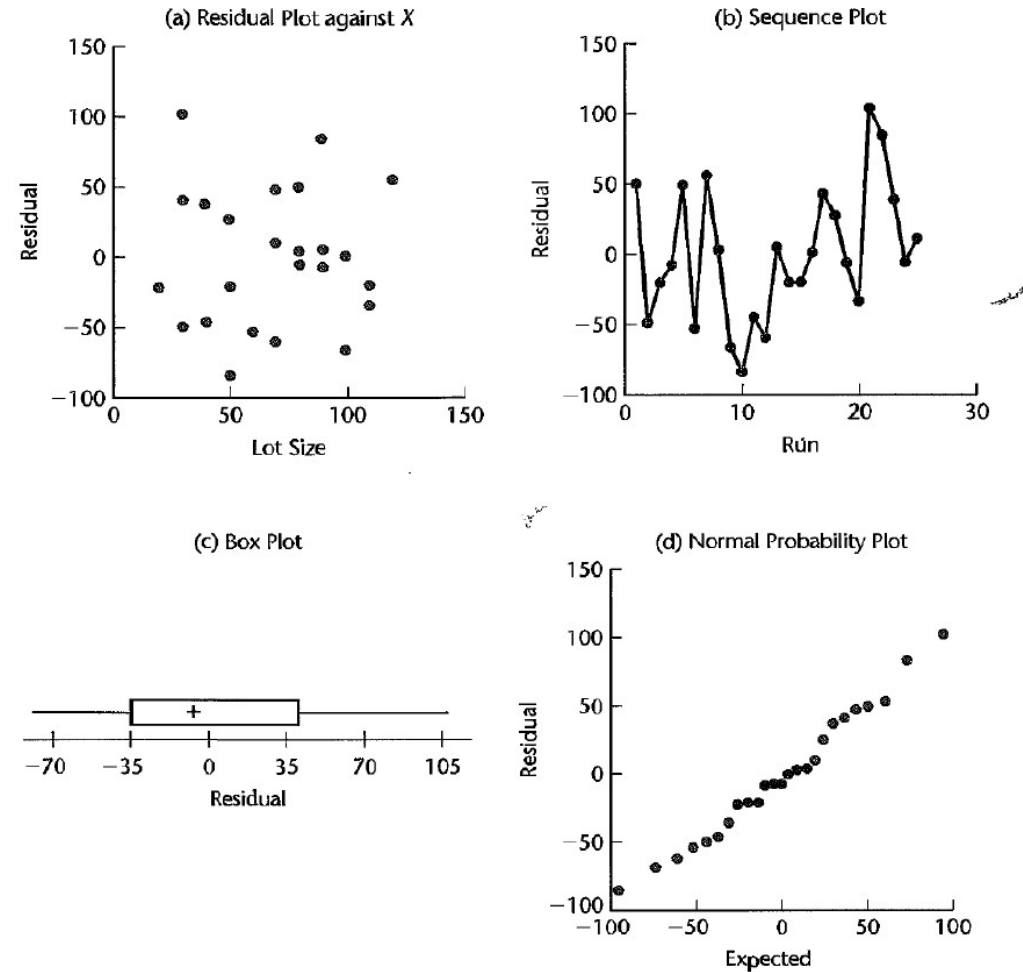


Figure : Diagnostic Residual Plots-Toluca Company Example.

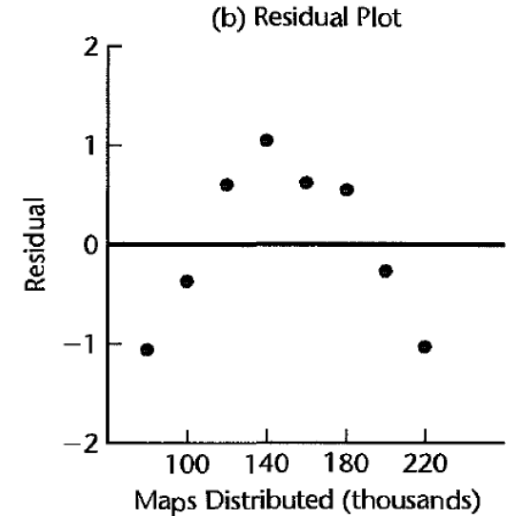
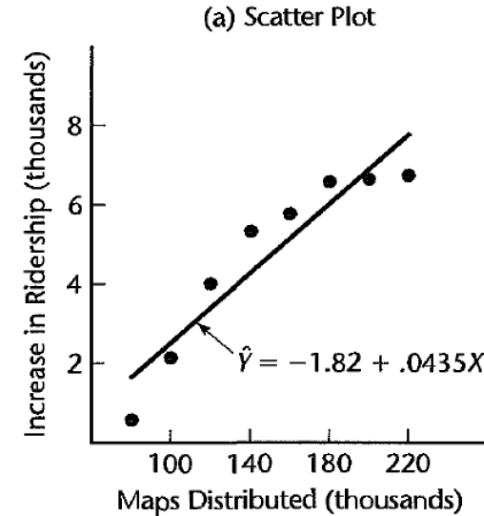
Nonlinearity of Regression Function

- Whether a linear regression function is appropriate for the data:
 - e_i vs. X
 - e_i vs. $\hat{Y}_i$
- Nonlinearity: can also be studied from a scatter plot (not always as effective as a residual plot)

Nonlinearity of Regression Function

	(1)	(2)	(3)	(4)
	Increase in Ridership (thousands)	Maps Distributed (thousands)	Fitted Value $\hat{Y}_i$	Residual $Y_i - \hat{Y}_i = e_i$
City i	Y_i	X_i		
1	.60	80	1.66	-1.06
2	6.70	220	7.75	-1.05
3	5.30	140	4.27	1.03
4	4.00	120	3.40	.60
5	6.55	180	6.01	.54
6	2.15	100	2.53	-.38
7	6.60	200	6.88	-.28
8	5.75	160	5.14	.61

$\hat{Y} = -1.82 + .0435X$

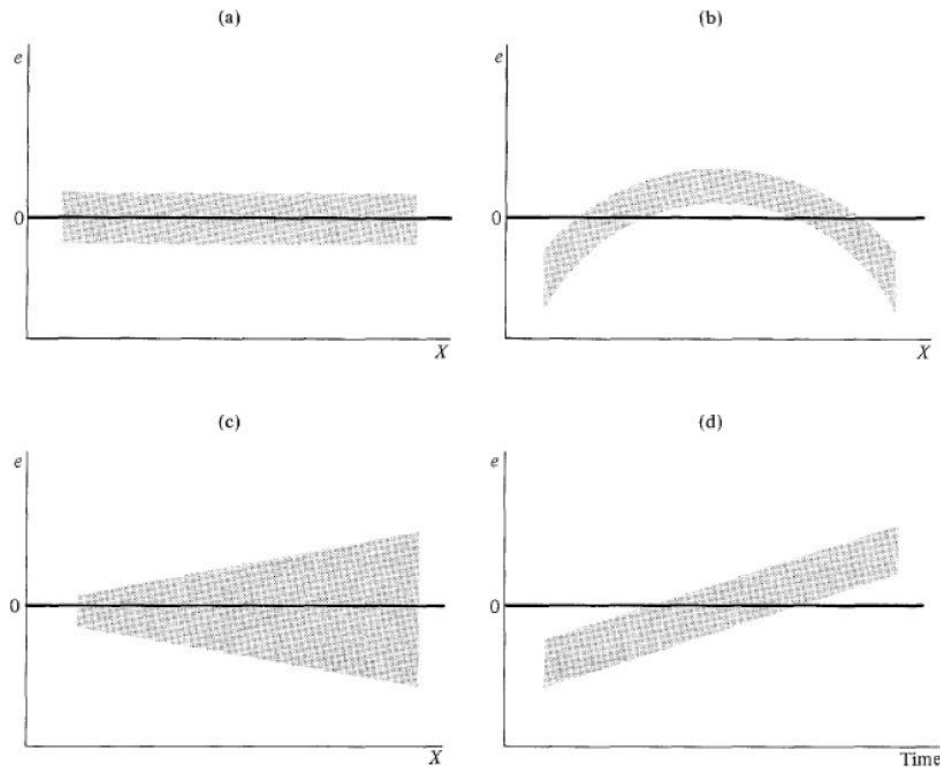


- The plot suggests strongly that a linear regression function is not appropriate.
- The residuals depart from 0 in a systematic fashion;
- The lack of fit of the linear regression function is even more strongly suggested by the residual plot against X than by the scatter plot.

Nonlinearity of Regression Function, cont'd

- The residual plot can easily be used for examining other factors of the aptness of the model.
- **Scaling problem:** becomes more difficult to study the appropriateness of a linear regression function from the scatter plot.
- Residual plot: clearly show systematic pattern

Nonlinearity of Regression Function, cont'd

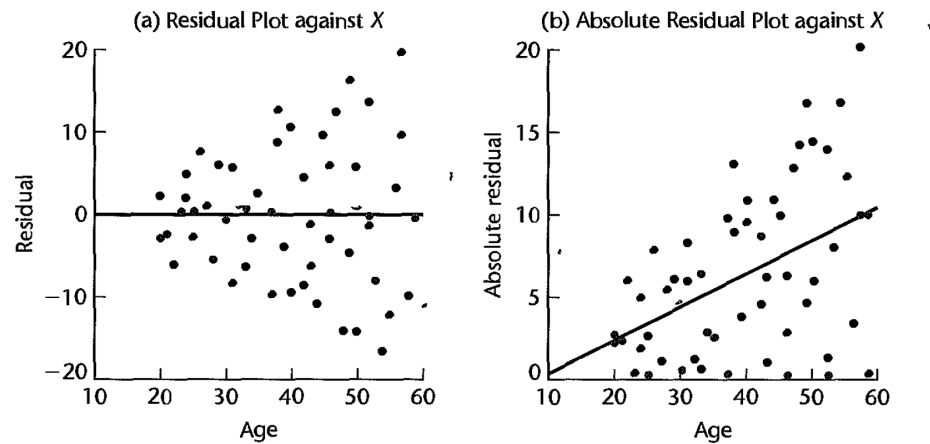


- (a) appropriate:
horizontal; around 0;
- (b) departure from the
linear regression
model: curvilinear

Figure : Prototype Residual Plots.

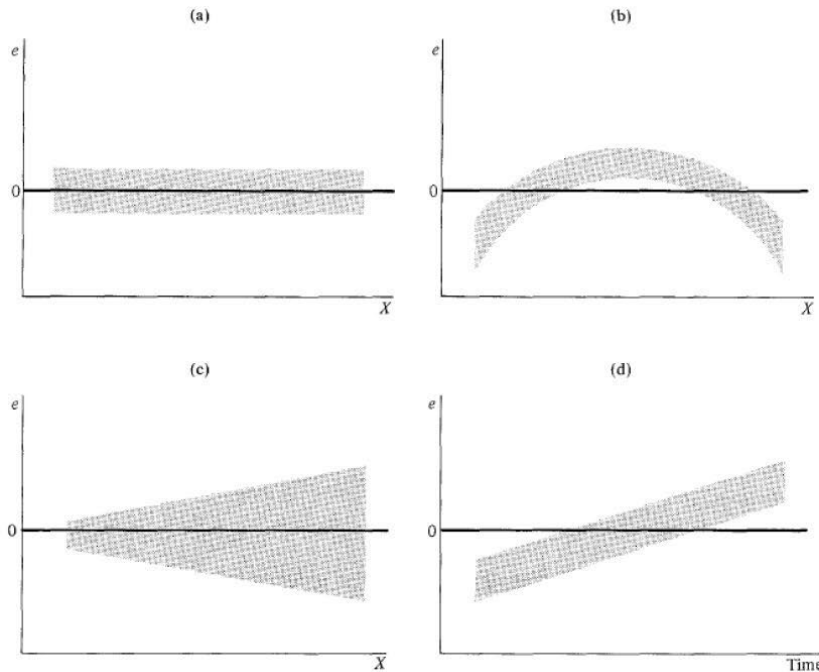
Nonconstancy of Error Variance

- Plot of e_i vs. X or $\hat{Y}_i$
 - study whether a linear regression function is appropriate
 - examine whether the variance of the error terms is constant



- Residual Plots Illustrating Nonconstant Error Variance.
- The signs of the residuals are not meaningful for examining the constancy of the error variance.

Nonconstancy of Error Variance, cont'd



- (a) constant variance
- (c) the error variance increases with X

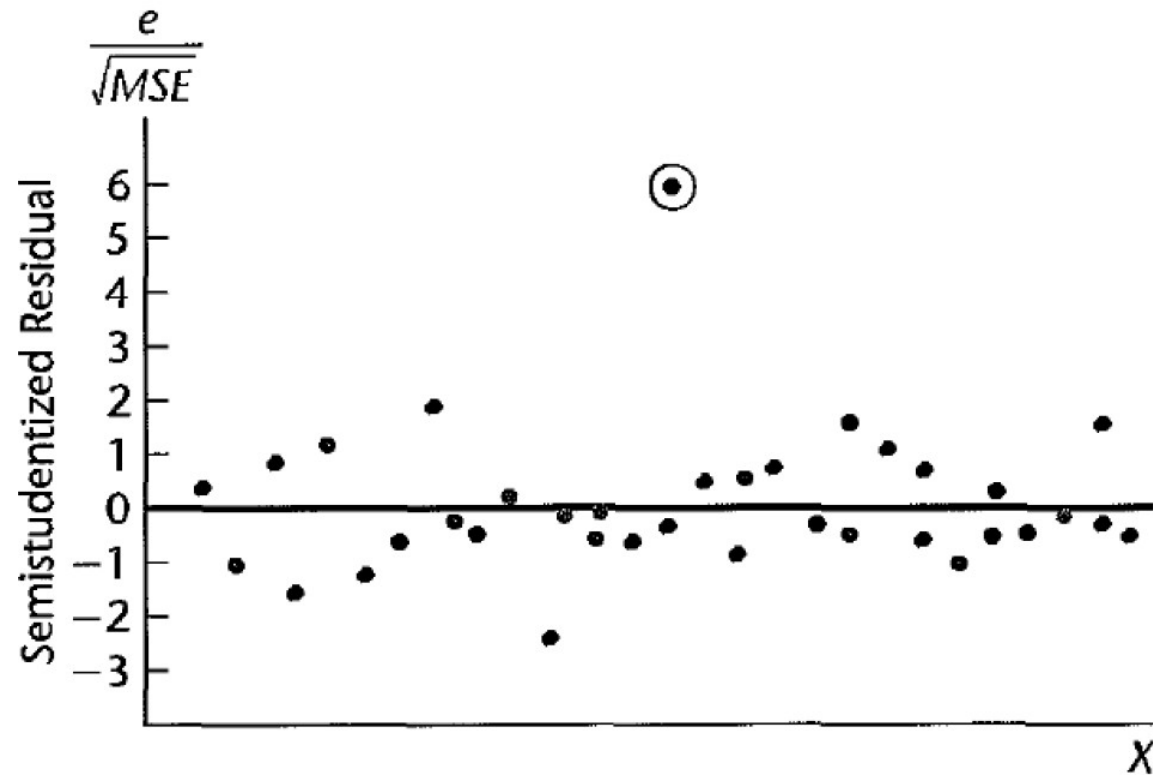
Figure : Prototype Residual Plots.

Presence of Outliers

- **Outliers:** extreme observations
- Residual outliers:
- identified from *residual plots* vs. X or $\hat{Y}_i$
 - box plots
 - stem-and-leaf plots
 - dot plots
- Semistudentized residuals:
 - helpful for distinguishing outlying observations
 - easy to identify residuals that lie many standard deviations from zero.

Presence of Outliers, cont'd

Figure : Residual Plot with Outlier.



Presence of Outliers, cont'd

Outliers can create great difficulty:

- Mistake → Discard
- Other extraneous effect → Discard
- A fitted line may be pulled disproportionately toward an outlier → cause a misleading fit
- **On the other hand, outliers may convey significant information**
- A safe rule frequently suggested is to discard an outlier only if there is direct evidence that it represents an error in recording, a miscalculation, a malfunctioning of equipment, or a similar type of circumstance.

Presence of Outliers, cont'd

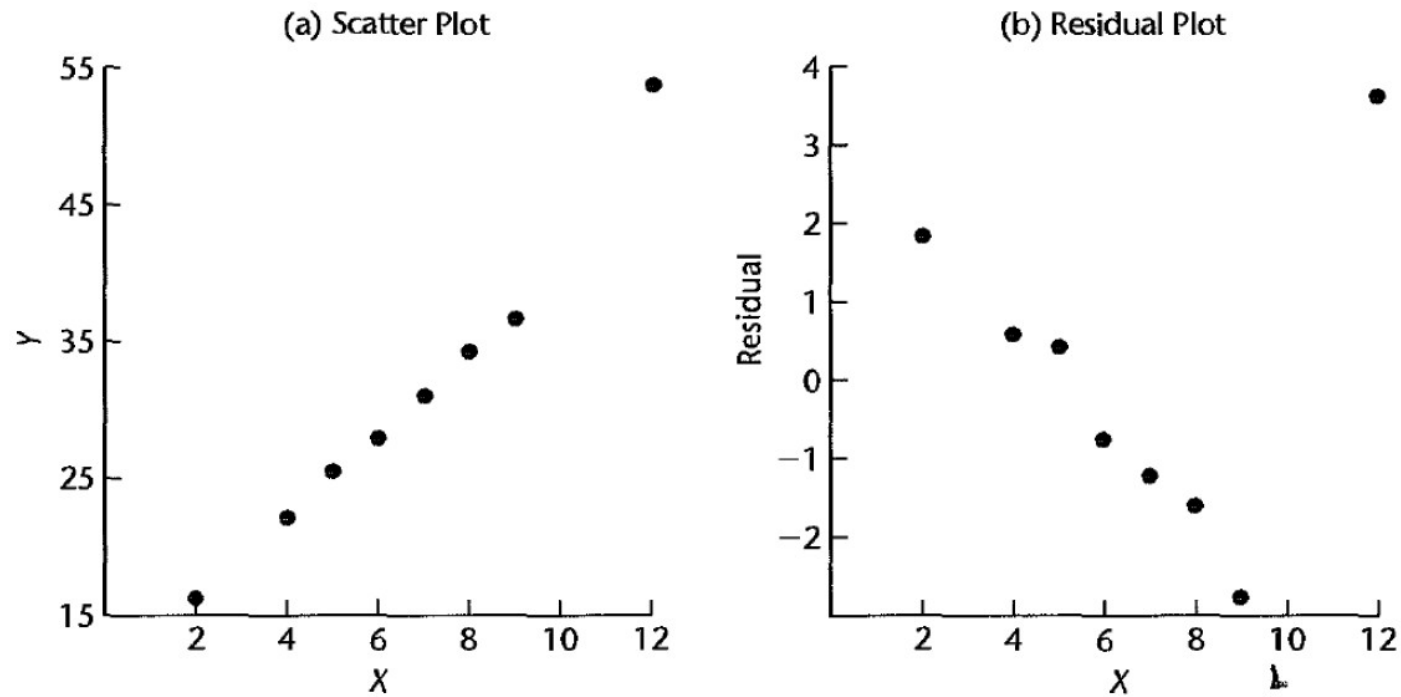
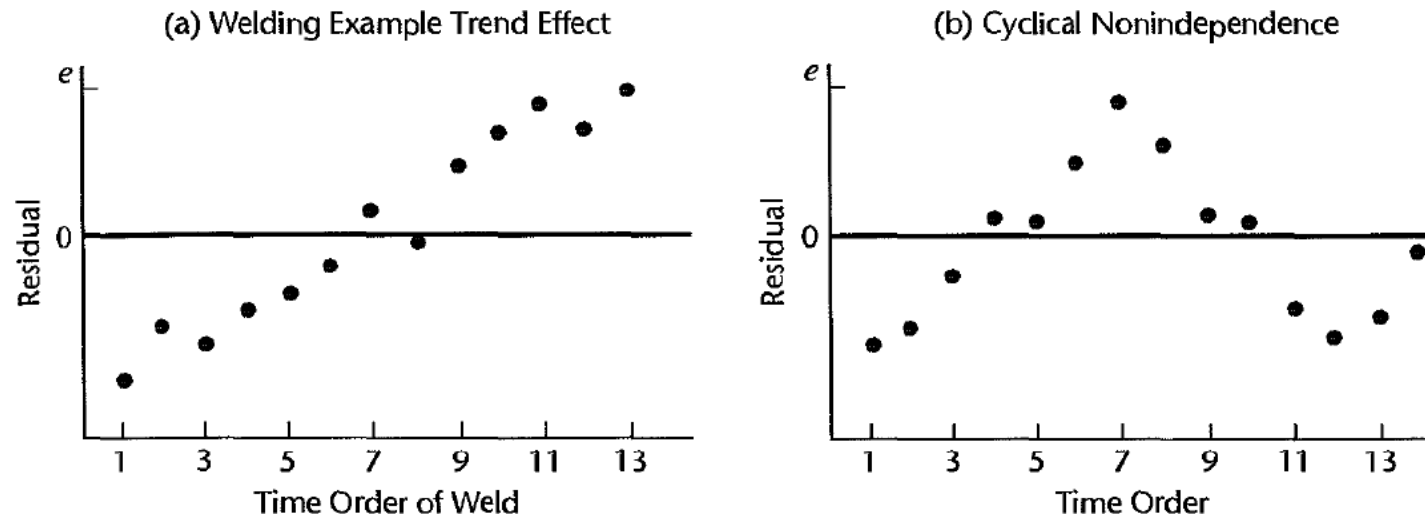


Figure : Distorting Effect on Residuals Caused by an Outlier When Remaining Data Follow Linear Regression.

Nonindependence of Error Terms

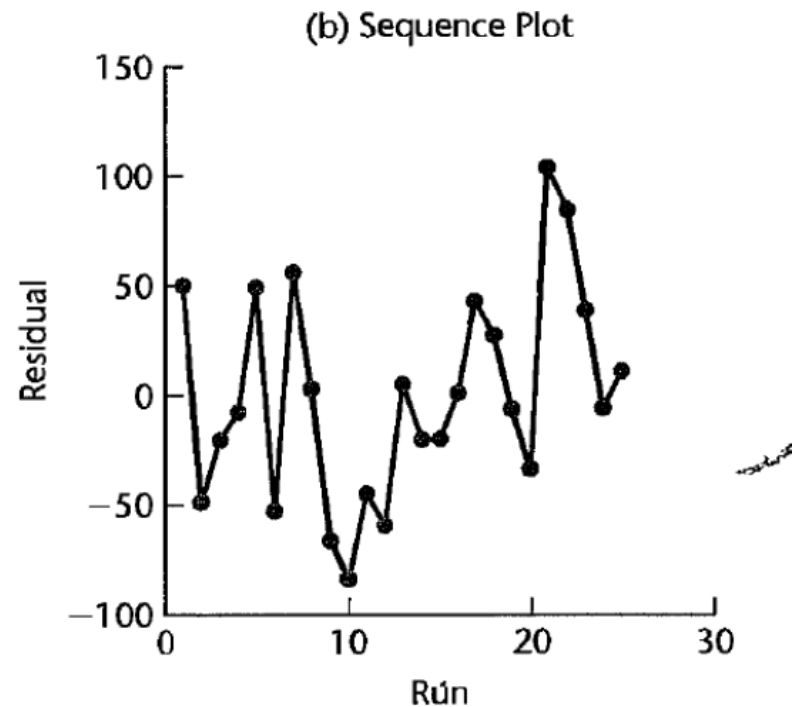
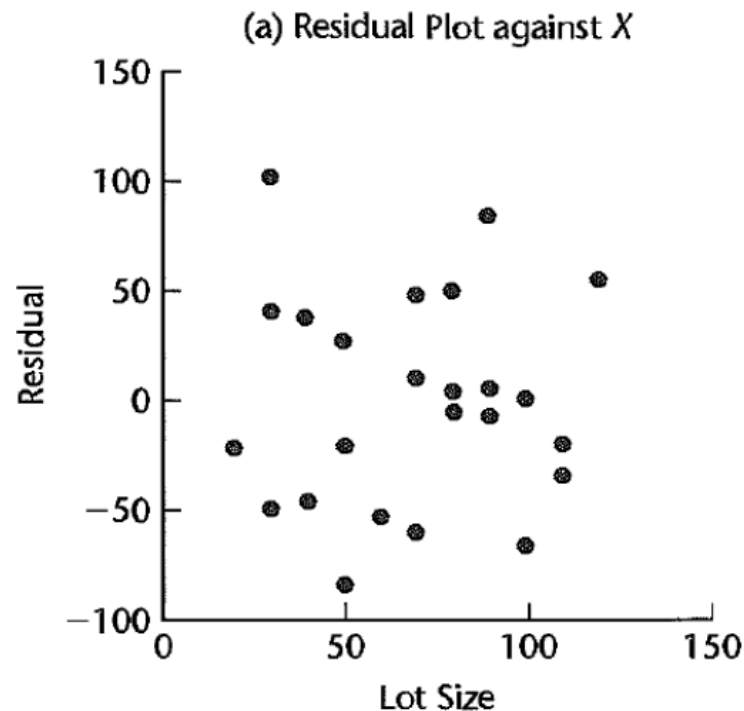
- A sequence plot of the residuals
- To see if there is any correlation between error terms



- Figure : Residual Time Sequence Plots Illustrating Nonindependence of Error Terms.
- Some effect connected with time was present.

Nonindependence of Error Terms, cont'd

- When the error terms are independent, we expect the residuals in a sequence plot to fluctuate in a more or less random pattern around the base line 0.



Nonnormality of Error

Terms

- Small departures from normality do not create any serious problems.
- Distribution Plots:
 - box plot: Symmetry
 - histogram; dot plot; stem-and-leaf plot
- Comparison of Frequencies: Compare actual frequencies of the residuals vs. expected frequencies under normality
 - about 68% of e_i fall between $\pm \sqrt{MSE}$
 - about 95% of e_i fall between $\pm 2 \sqrt{MSE}$

Nonnormality of Error Terms, cont'd

- **Normal Probability Plot:**
 - Each residual e_i is plotted vs. its expected value under normality.
 - Nearly linear suggests agreement with normality
- Table : Residuals and Expected Value under Normality- Toluca

Compan	(1)	(2)	(3)
Run i	Residual e_i	Rank k	Expected Value under Normality
1	51.02	22	51.95
2	-48.47	5	-44.10
3	-19.88	10	-14.76
...	...	...	...
23	38.83	19	31.05
24	-5.98	13	0
25	10.72	17	19.93

Nonnormality of Error Terms, cont'd

- The expect value of the error terms for is 0.
- The standard deviation for the error terms: $\sqrt{MSE}$

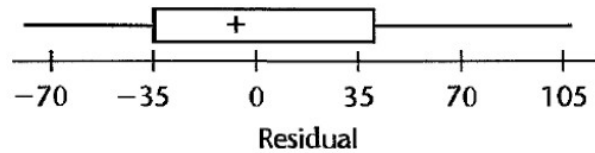
$$\Rightarrow \sqrt{MSE} \left[z \left(\frac{k - .375}{n + .25} \right) \right]$$

$z(A)$ the $(A)100$ percentile of $N(0, 1)$

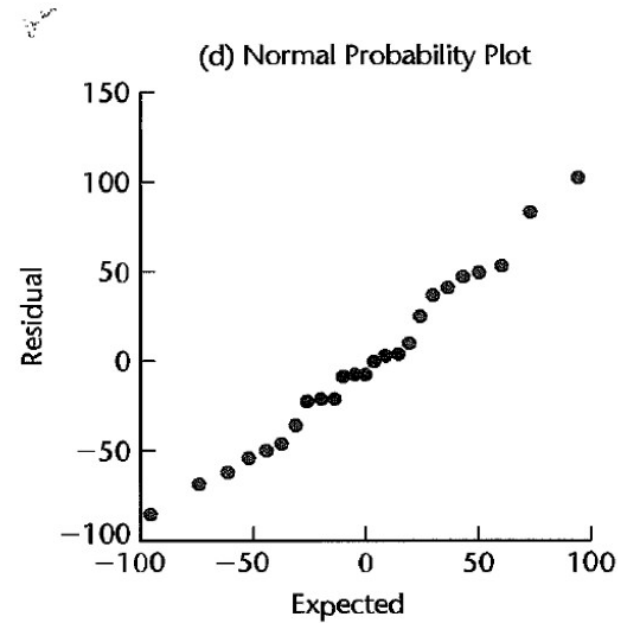
Nonnormality of Error Terms, cont'd

$e_1 = 51.02$ vs. the expected value $\sqrt{2384}[z(.8564)] = 51.95$

(c) Box Plot



(d) Normal Probability Plot



Nonnormality of Error Terms, cont'd

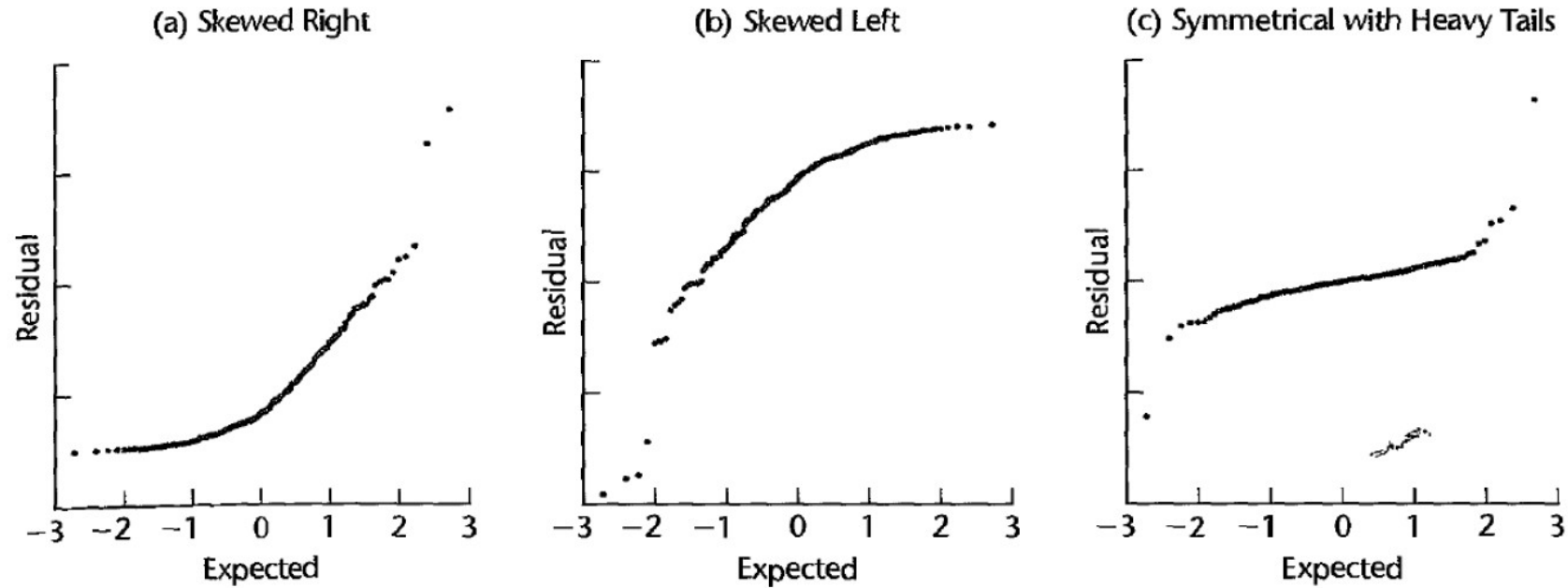


Figure : Normal Probability Plots when Error Term Distribution Is Not Normal.

Omission of Important Predictor

Variables omitted from the model

- To determine whether there are any key variables that could provide important additional descriptive and predictive power to the model

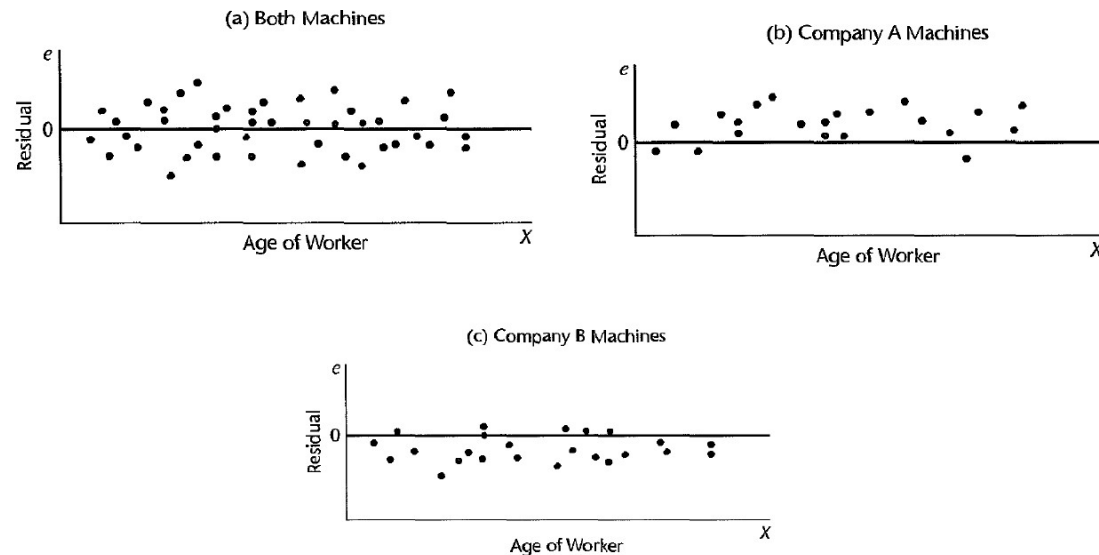


Figure : Residual Plots for Possible Omission of Important Predictor Variable-Productivity Example.

Overview of Tests Involving Residuals

- Tests for Randomness: Run test; Durbin-Watson Test
- Tests for Constant Variance: residual plot; Brown-Forsythe test; Breusch-Pagan test
- Tests for Outliers: Multiple tests will be explained later
- Tests for Normality: Goodness of fit tests; chi-square test; K-S test; etc.

Correlation Test for Normality

- No test is necessary. This can be visually detected from QQ plot.
- However, Shapiro-Wilk test can be performed:

Ho: Errors are normally distributed

Ha: Errors are Not normally distributed

```
> shapiro.test(toluca.reg$residuals)
```

```
Shapiro-Wilk normality test
```

```
data: toluca.reg$residuals  
W = 0.9789, p-value = 0.8626
```

- P-Value>0.05 accept the Ho. Errors are normally distributed

Tests for Constancy of Error Variance

Two formal tests: Do the error terms have constant variance?

1. Brown-Forsythe Test
2. Breusch-Pagan test

Brown-Forsythe Test

- A modification of the Levene test, does not depend on normality of the error terms
- Robust against serious departures from normality
- relative efficient when the error terms are normally distributed
- The Brown-Forsythe test is applicable to simple linear regression when the variance of the error terms either increases or decreases with X .
- The sample size needs to be large enough so that the dependencies among the residuals can be ignored.

Brown-Forsythe Test, cont'd

- The test is based on the variability of the residuals.
- The larger the error variance, the larger the variability of the residuals will tend to be.
- Divide the data set into two groups, according to X **low** vs. **high**
- Robust consideration: utilize $|e_{ij} - \text{medians}(e_{ij})|$, $j = 1, 2$

Brown-Forsythe Test, cont'd

The Brown-Forsythe test consists simply of the two-sample t test based on test statistic

$$t^* = \frac{\bar{Y} - \bar{Z}}{s\{\bar{Y} - \bar{Z}\}} \quad (\text{A.67})$$

to determine whether the mean of the absolute deviations from one group differs significantly from the one for the second group.

Brown-Forsythe Test, cont'd

- e_{ij} : the i th residual for group j ; $j = 1, 2$.
- n_j , $j = 1, 2$: the sample sizes of the two groups; $n = n_1 + n_2$
- $\tilde{e}_j$: the medians of the residuals in the two groups
- $d_{ij} = |e_{ij} - \tilde{e}_j|$; $\bar{d}_j$: the sample means of the d_{ij}
- The Brown-Forsythe test:

$$t_{BF}^* = \frac{\bar{d}_1 - \bar{d}_2}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t(n-2) \quad \text{if the error terms have constant variance} \quad (1)$$

$$s^2 = \frac{\sum (d_{i1} - \bar{d}_1)^2 + \sum (d_{i2} - \bar{d}_2)^2}{n - 2} \quad (\text{the pooled variance}) \quad (2)$$

Brown-Forsythe Test, cont'd

Table : Brown-Forsythe Test (Constancy of error Variance): Toluca Company Example.

Group 1						Group 2					
		(1)	(2)	(3)	(4)			(1)	(2)	(3)	(4)
		Lot	Residual					Lot	Residual		
i	run	Size	e_{i1}	d_{i1}	$(d_{i1} - \bar{d}_1)^2$	i	run	Size	e_{i2}	d_{i2}	$(d_{i2} - \bar{d}_2)^2$
1	14	20	-20.77	.89	1,929.41	1	1	80	51.02	53.70	637.56
2	2	30	-48.47	28.59	263.25	2	8	80	4.02	6.70	437.06
...	...	...	...	...	...	...	...	...	...	...	...
12	12	70	-60.28	40.40	19.49	11	20	110	-34.09	31.41	8.76
13	25	70	10.72	30.60	202.07	12	7	120	55.21	57.89	866.71
Total				582.60	12,566.6	Total				341.40	9,610.2
		$\tilde{e}_1 = -19.88$	$\bar{d}_1 = 44.815$					$\tilde{e}_2 = -2.68$	$\bar{d}_2 = 28.450$		

Since $|t_{BF}^*| \leq 2.069$ ($\alpha = 0.05$) $\Rightarrow$ the error variance is constant and does not vary with the level of X (two-sided P -value=0.20)

R Code

```
install.packages("onewaytests")
```

```
library(onewaytests)
```

```
bf.test(formula, data, alpha = 0.05)
```

```
bf.data<-data.frame(cbind(toluca_data,toluca.reg$residuals,toluca.reg$fitted.values))
```

```
> dimnames(bf.data)[[2]][3]
```

```
[1] "toluca.reg.residuals"
```

```
> dimnames(bf.data)[[2]][3:4]
```

```
[1] "toluca.reg.residuals"      "toluca.reg.fitted.values"
```

```
> dimnames(bf.data)[[2]][3:4]<-c("residuals","fitted.values")
```

```
bf.data1<-data.frame(cbind(bf.data,ind=as.factor(I(bf.data$lotsize<=median(bf.data$lotsize))*1)))
```

```
> bf.test(residuals~ind,data=bf.data1)
```

Brown-Forsythe Test ($\alpha = 0.05$)

```
data : residuals and ind
```

```
statistic      : 0.2467212
```

```
num df      : 1
```

```
denom df      : 22.17293
```

```
p.value      : 0.6242822
```


Breusch-Pagan Test

A large-sample test: assume that the error terms are independent and normally distributed and $\sigma^2\{\varepsilon_i\} = \sigma_i^2$ is related to X

$$\log_e \sigma_i^2 = \gamma_0 + \gamma_1 X_i$$

- σ_i^2 increase or decrease with X , depending on the sign of γ_1
- Constancy of error variance $\Rightarrow \gamma_1=0$

Breusch-Pagan Test.

- Test:

$$H_0 : \gamma_1 = 0 \text{ vs. } H_a : \gamma_1 \neq 0$$

by e_i^2 vs. X_i .

- The test statistic:

$$X_{BP}^2 = \frac{SSR^*}{2} \div \left(\frac{SSE}{n}\right)^2 \sim \chi^2(1) \quad n \text{ is reasonably large}$$

- Example: Toluca Company
- $SSR = 7, 896, 128$ and $SSE = 54,825$
- $0.821 \leq 3.84 = \chi^2(0.95; 1)$ conclude H_0 : the error variance is constant

Overview of Remedial Measures

If the simple linear regression model is not appropriate:

- Abandon regression model
 - better insights
 - more complex procedures for estimating the parameters
- Employ some transformation
 - relatively simple methods of estimation
 - Transformation may be complex

Transformations

- Transformations for Nonlinear Relation Only
- Transformations for Nonnormality and Unequal Error Variances
- Box-Cox Transformations
- Transformations for one or both of the original variables before carrying out the regression analysis
- To make the regression model appropriate for the transformed data
 - Several alternative transformations may be tried
 - Scatter plot and residual plots based on each transformation should then be prepared and analyzed to decide which transformation is most effective.

Transformation, cont'd

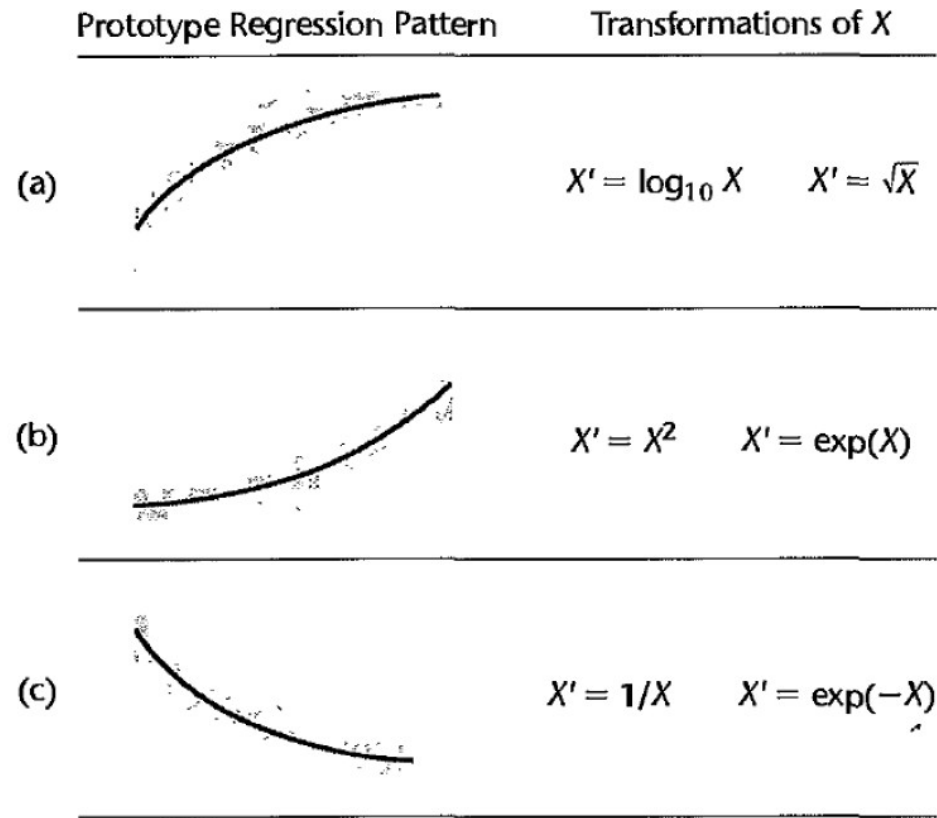
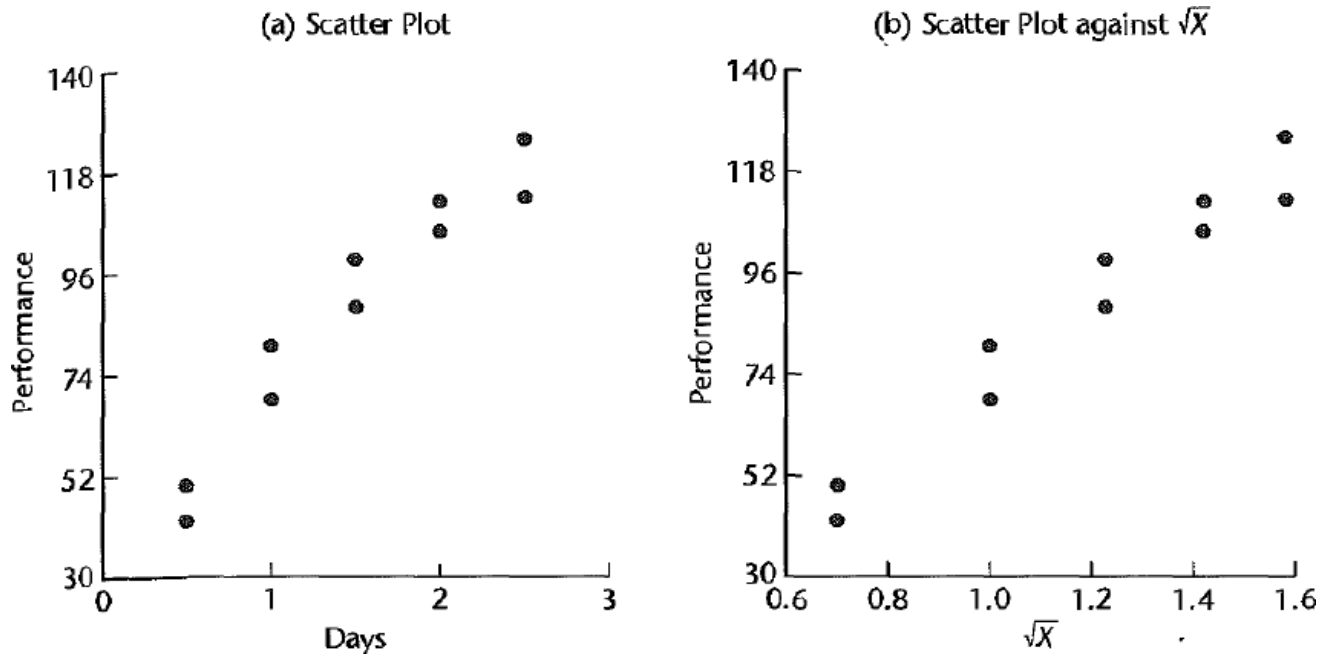


Figure : Prototype Nonlinear Regression Patterns with Constant Error Variance and Simple Transformations of X .

Transformation, cont'd

Example: 10 participants in the study

- X : the number of days of training received
- Y : performance score in a battery of simulated sales situations



Transformation, cont'd

Table : Use of Square Root Transformation of X to Linearize Regression Relation Sales Training Example.

	(1)	(2)	(3)
Sales Trainee	Days of Training	Performance Score	
i	X_i	Y_i	$X'_i = \sqrt{X_i}$
1	.5	42.5	.70711
2	.5	50.6	.70711
3	1.0	68.5	1.00000
4	1.0	80.7	1.00000
5	1.5	89.0	1.22474
6	1.5	99.6	1.22474
7	2.0	105.3	1.41421
8	2.0	111.8	1.41421
9	2.5	112.3	1.58114
10	2.5	125.7	1.58114

Transformation, cont'd

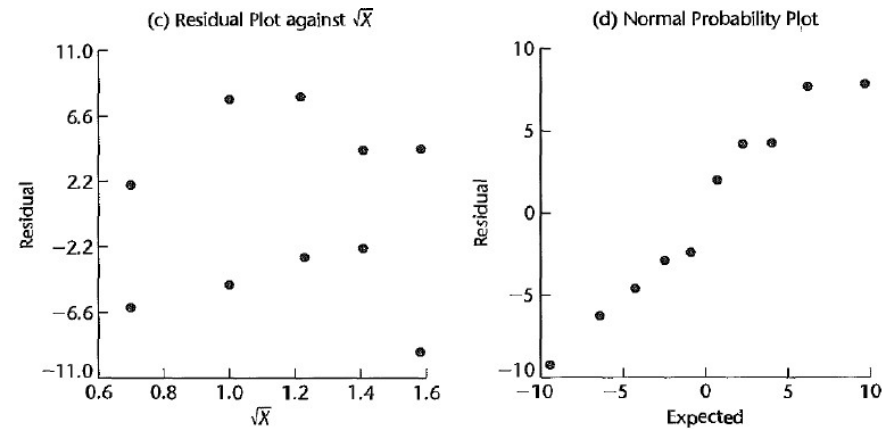


Figure : Scatter Plots and Residual Plots-Sales Training Example.

- Fitted regression function: $\hat{Y} = -10.33 + 83.45X'$
- There is no evidence of lack of fit or of strongly unequal error variances.

Transformation, cont'd

- Unequal error variances; nonnormality of the error terms: transformation on Y
- also helps to linearize a curvilinear regression relation
- A simultaneous transformation on X may also be helpful or necessary.
- Ex: Y : household expenditures for vacations; X : household income
 - more variation
 - greater positive skewness

Transformation, cont'd

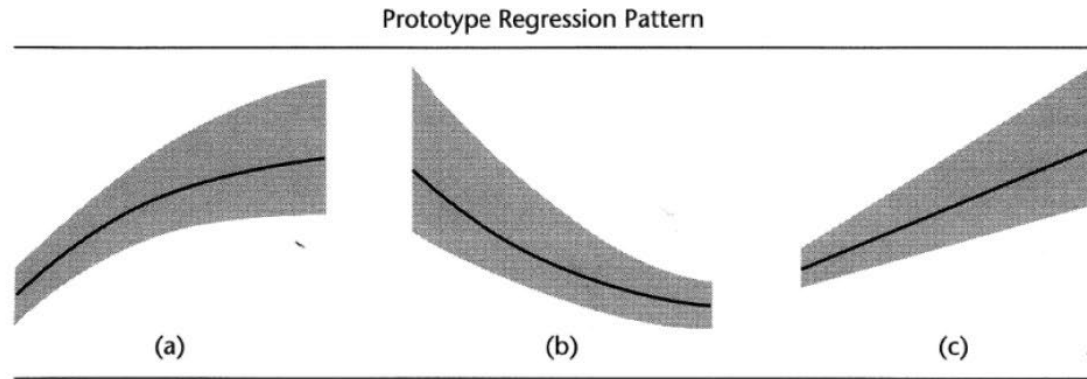


Figure : Prototype Regression Patterns with Unequal Error Variances and Simple Transformations of Y .

Transformation on Y

$$Y' = \sqrt{Y}$$

$$Y' = \log_{10} Y$$

$$Y' = 1/Y$$

Transformation, cont'd

Table : Use of Logarithmic Transformation of Y to Linearize Regression Relation and Stabilize Error Variance-Plasma Levels Example.

Child i	(1) Age X_i	(2) Plasma Level Y_i	(3) $Y'_i = \log_{10} Y_i$
1	0 (newborn)	13.44	1.1284
2	0 (newborn)	12.84	1.1086
3	0 (newborn)	11.91	1.0759
4	0 (newborn)	20.09	1.3030
5	0 (newborn)	15.60	1.1931
6	1.0	10.11	1.0048
7	1.0	11.38	1.0561
...	...	...	...
19	3.0	6.90	.8388
20	3.0	6.77	.8306
21	4.0	4.86	.6866
22	4.0	5.10	.7076
23	4.0	5.67	.7536
24	4.0	5.75	.7597
25	4.0	6.23	.7945

Transformation, cont'd

$Y' = \log_{10} Y$: $\hat{Y}' = 1.135 - 0.1023X$; The coefficient of correlation = $0.981 > 0.959$

linear relation; variability at the different X level

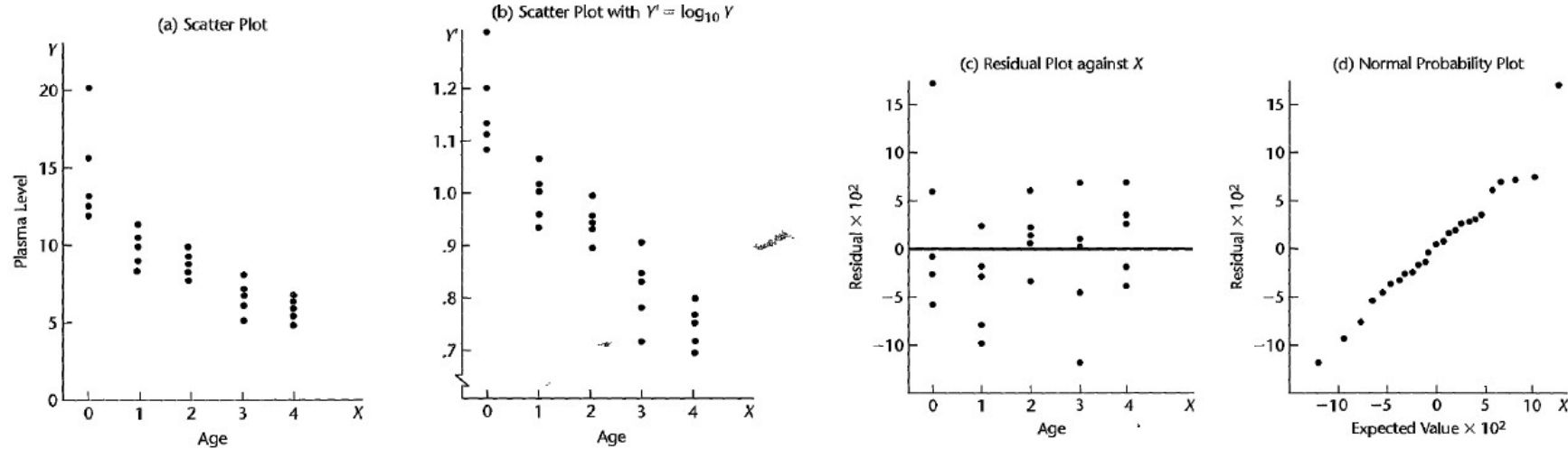


Figure : Scatter Plots and Residual Plots-Plasma Level Example.

Box-Cox Transformation

- It is often difficult to determine from diagnostic plots which transformation of Y is most appropriate
 - correcting skewness of the distributions of error terms
 - unequal error variances
 - nonlinearity of the regression function
- Box-Cox procedure: automatically identifies a transformation from the family of power transformations on Y

Box-Cox Transformation, cont'd

The family of power transformations:

$$Y' = Y^\lambda$$

λ : a parameter to be determined from the data

$$\lambda = 2 \qquad Y' = Y^2$$

$$\lambda = .5 \qquad Y' = \sqrt{Y}$$

$$\lambda = 0 \qquad Y' = \log_e Y \quad (\text{by definition})$$

$$\lambda = -.5 \qquad Y' = \frac{1}{\sqrt{Y}}$$

$$\lambda = -1.0 \qquad Y' = \frac{1}{Y}$$

Box-Cox Transformation, cont'd

The model becomes:

$$Y_i^\lambda = \beta_0 + \beta_1 X_i + \varepsilon_i$$

λ : needs to be estimated by MLE as well as $\beta_0, \beta_1, \sigma^2$

Simple procedure to obtain $\hat{\lambda}$: search in a range of potential λ

- λ : grid on $-2, -1.75, \dots, 1.75, 2$
- Each λ : standardized the Y_i^λ ;
- The magnitude of the error sum of squares does not depend on λ :

Box-Cox Transformation, cont'd

$$W_i = \begin{cases} K_1(Y_i^\lambda - 1) & \lambda \neq 0 \\ K_2(\log_e Y_i) & \lambda = 0 \end{cases}$$

$$K_2 = \left(\prod_{i=1}^n Y_i \right)^{\frac{1}{n}} ; \quad K_1 = \frac{1}{\lambda K_2^{\lambda-1}}$$

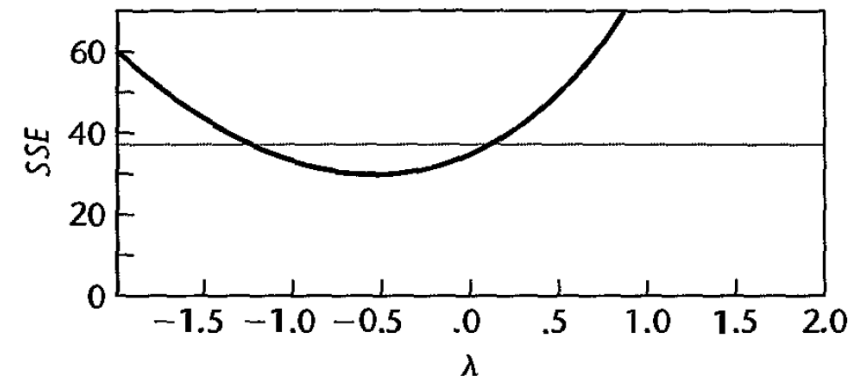
The MLE $\hat{\lambda}$ is that value of λ for which SSE is a minimum.

Scatter and residual plots should be utilized to examine the appropriateness of the transformation identified by the Box-Cox procedure.

Box-Cox Transformation, cont'd

- Plasma Level example:
- SSE is the lowest, when $\lambda = -0.5$

λ	SSE	λ	SSE
1.0	78.0	-.1	33.1
.9	70.4	-.3	31.2
.7	57.8	-.4	30.7
.5	48.4	-.5	30.6
.3	41.4	-.6	30.7
.1	36.4	-.7	31.1
0	34.5	-.9	32.7
		-1.0	33.9



Box-Cox Transformation, cont'd

- Theoretical or a priori considerations can be utilized to help in choosing an appropriate transformation.
- When transformation models are employed, the estimators b_0, b_1 have the least squares properties with respect to the transformation observations, not the original ones.
- The MLE of λ with the Box-Cox procedure is subject to sampling variability.
 - SSE is fairly stable in a neighborhood around the estimate
 - a nearby λ : easy to understand; Ex: $\lambda = 0$ instead of $\hat{\lambda} = 0.13$
- λ near 1, no transformation of Y is needed

Exploration of Shape of Regression Function

- If the scatter plot is complex, difficult to see the nature of the regression relationship
- fitting a smoothed curve without any constraints on the regression function: **nonparametric regression curve**
 - exploring regression relationship
 - confirming the nature of the regression function

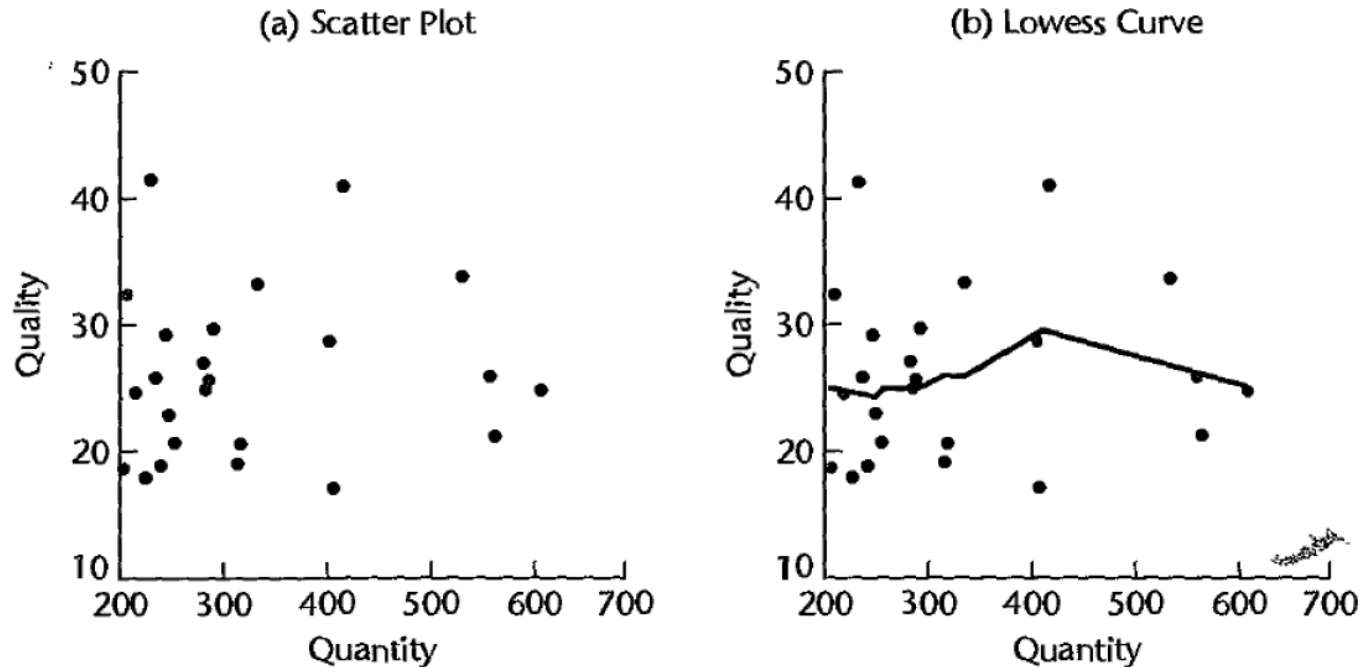
Exploration of Shape of Regression Function, cont'd

- Smoothing methods:
 - Method of moving average: time series data
 - Method of running medians: similar to the method of moving average
 - Band regression
- Band regression:
 - Divided the data into a number of groups or "bands"
 - Median of X and Y
 - connected by straight lines

Exploration of Shape of Regression Function, cont'd

- Lowess Method: more refined nonparametric method than band regression
 - fitting successive linear regression functions in local neighborhoods to obtain a smooth curve
 - Lowess: locally weighted regression scatter plot smoothing

Exploration of Shape of Regression Function, cont'd



- Figure : Scatter Plot and Lowess Smoothed Curve-Research Laboratories Example.

Exploration of Shape of Regression Function, cont'd

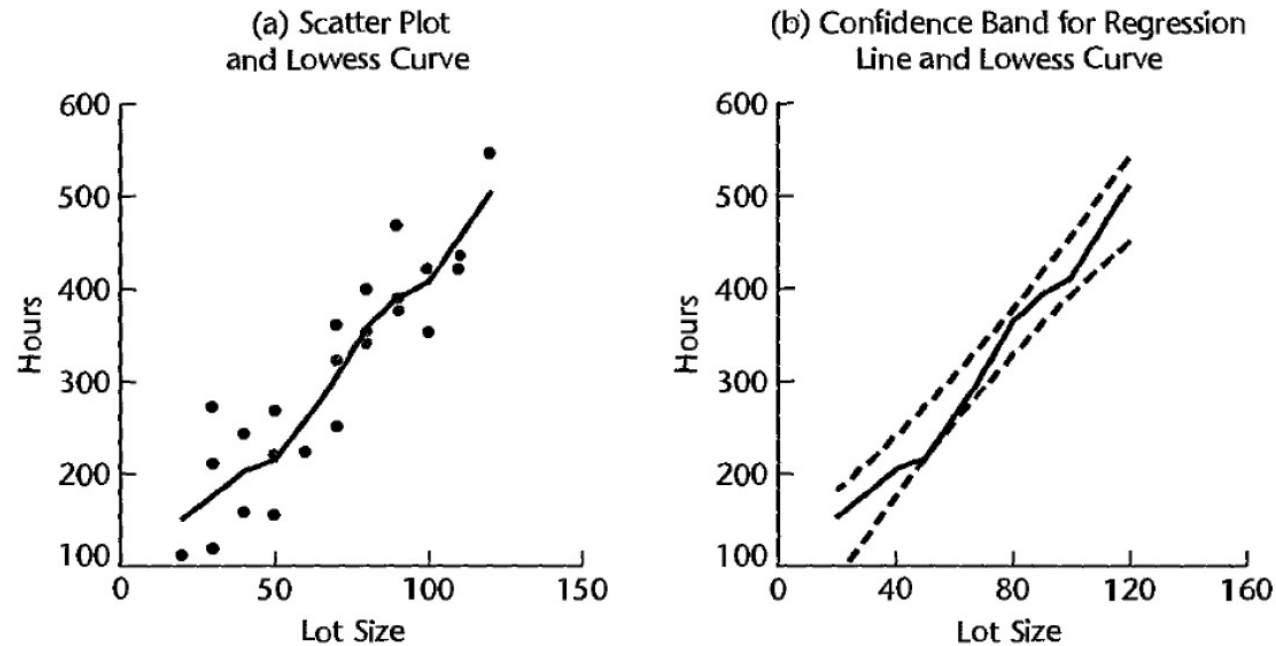


Figure : MINTAB Lowess Curve and Confidence Band for Regression Line-Toluca Company Example.

Exploration of Shape of Regression Function, cont'd

- Smoothed curves do not provide an analytical expression for the function form of the regression relationship.
- Only suggest the shape of the regression curve
- The lowess procedure is not restricted to fitting the regression function in each neighborhood. (also Higher-degree polynomials)